

Labour Supply and Public Policy

ECON 300 - Labour Economics

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Learning Objectives

- Discuss key features of demogrants, social assistance, unemployment insurance, wage subsidies, workers' compensation, and childcare subsidies.
- Use the labour supply model to analyze work incentives under demogrants, social assistance (with clawbacks), and wage subsidies.
- Analyze unemployment insurance within the income–leisure model and note empirical challenges.
- Graphically illustrate disability impacts on the budget constraint; compare compensation programs.
- Show how childcare costs and subsidies alter constraints and participation decisions.

Main Questions

- Can Canadian transfer programs be analyzed in a common framework to assess “best” designs for work incentives?
- Does unemployment insurance necessarily reduce incentives to work?
- Can welfare be designed to minimize adverse incentives while targeting need?
- How do we incorporate disabilities into the labour supply model and evaluate workers' compensation?
- How do childcare costs and subsidies affect participation, hours, and the reservation wage?

Income Maintenance Programs

- Aim to supplement low incomes that arise for varied reasons; no single program fits all causes.
- Distinguish **permanent** (low wage) from **transitory** (hours shock) shortfalls—policy tools differ.
- **Universal** vs. **Targeted**: simplicity and coverage vs. cost and incentive effects.

Universal vs. Targeted Programs

Universal Programs

Administratively simple; everyone receives the same transfer regardless of income. Expensive and benefits also flow to non-poor.

Targeted Programs

Cheaper: transfers top up to a threshold (e.g., poverty line) but can create **implicit taxes** on earnings that reduce work effort.

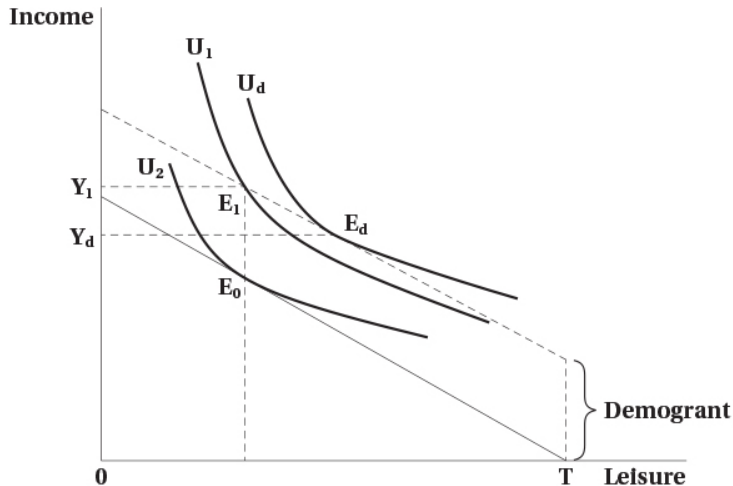
TABLE 3.1: Government transfers received by Census Families in Canada, 2017

Benefits	Percent of families	Average Amount
All government transfers (federal and provincial)	83.7%	\$12,750
Federal Child Benefits	20.2%	\$6,314
Provincial Child Benefits	10.6%	\$1,941
Employment Insurance	14.0%	\$8,352
Social Assistance	11.3%	\$8,367
Workers' Compensation	3.5%	\$9,820
Canada/Quebec Pension Plan	32.4%	\$10,092
Old Age Security	25.1%	\$8,566
Guaranteed Income Supplement	9.6%	\$6,634

- Lump-sum transfer specific to a demographic group (e.g., OAS).
- **Work incentives:** No substitution effect; a pure **income effect** that tends to increase leisure and reduce hours.

Note: The income gain is partly “spent” on leisure, so income rises by less than the grant amount.

Figure 3.1: Work Incentive Effects of a Lump-Sum Demogrant



Welfare (Social Assistance)

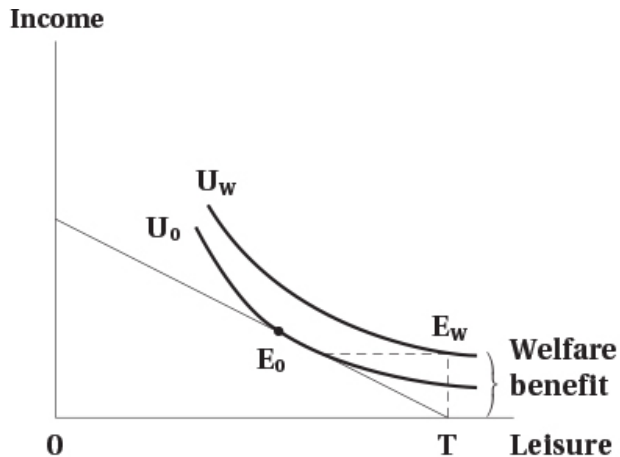
- Administered by provinces; partly financed by federal government.
- Benefits depend on need, assets, and other incomes.

TABLE 3.2: Income of Welfare Recipients by Province, 2018 (lone parent, one child)

Province	Annual Income ^a (\$)	Income as % of Poverty Line ^b
Newfoundland	23,436	85
Prince Edward Island	20,977	77
Nova Scotia	18,240	67
New Brunswick	19,978	78
Quebec	21,867	86
Ontario	21,463	72
Manitoba	21,764	82
Saskatchewan	21,087	77
Alberta	19,927	68
British Columbia	20,782	71
Average (unweighted)	20,952	76

a. Total annual income under program rules. *b.* Relative to poverty threshold used in the chapter.

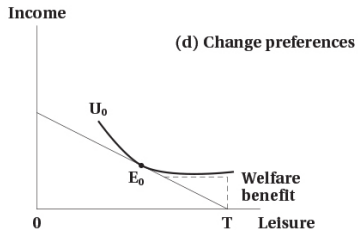
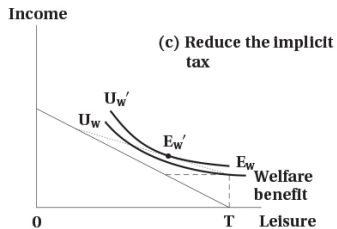
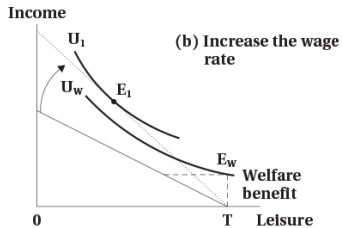
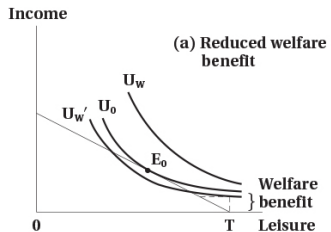
Figure 3.2: Work Incentive Effects of a Welfare Benefit with 100% Clawback



Improving Work Incentives in Welfare

- Reduce the welfare benefit level (lower income effect).
- Raise the wage rate (tilt budget upward).
- Reduce the implicit tax rate (partial clawback).
- Shift preferences (e.g., work supports) toward market work.

Figure 3.3: Other Work Incentive Effects of Welfare Programs

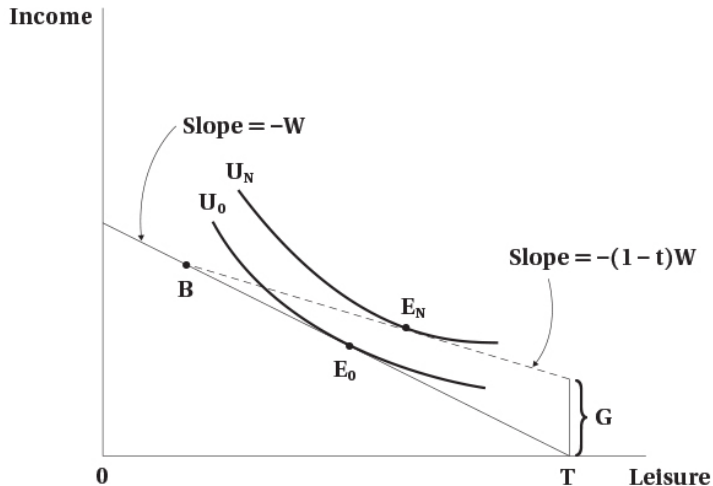


Income guarantee with partial clawback:

$$Y = G + (1 - t)E$$

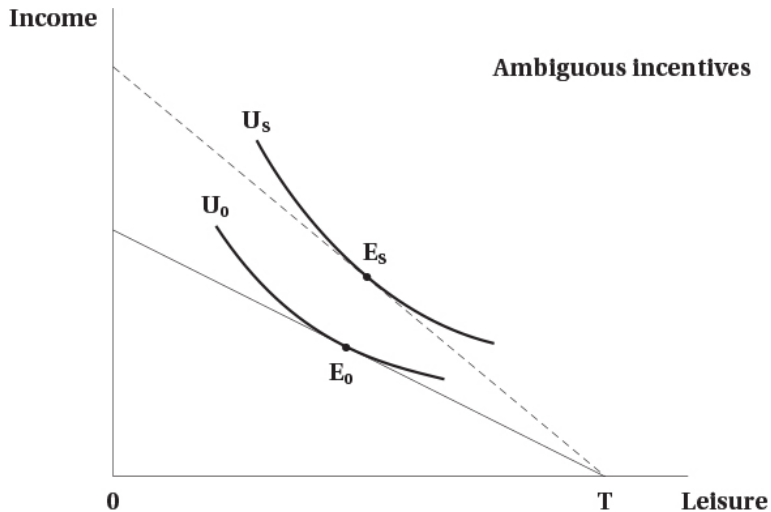
- G : guaranteed income; t : implicit tax rate; E : earnings.
- Reduces hours via income effect and partial substitution effect (slope changes).
- Can replace welfare schemes with harsher disincentives.

Figure 3.5: Work Incentive Effects of a Negative Income Tax



- Subsidizes earnings; raises effective wage.
- **Theoretical net effect on hours is indeterminate** (income vs. substitution work in opposite directions).
- Less adverse than a pure NIT for workers but does not help those unable to work.

Figure 3.6: Work Incentive Effects of a Wage Subsidy

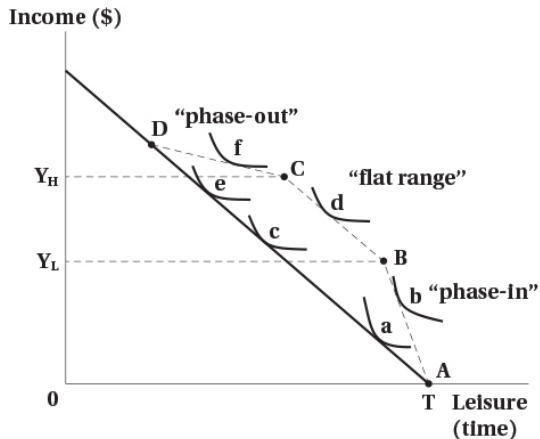


Refundable Tax Credit (Targeted Wage Subsidy)

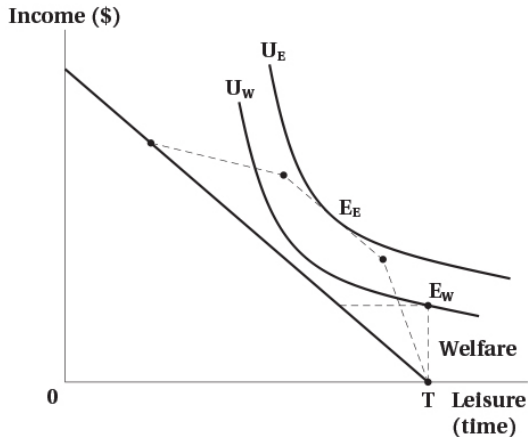
- Income-tested wage subsidy: **phase-in**, **flat**, and **phase-out** regions.
- Phase-in: substitution $\uparrow$ (work more) vs. income $\uparrow$ (work less) $\Rightarrow$ ambiguous net.
- Flat: income effect only $\Rightarrow$ hours fall (if leisure is normal).
- Phase-out: acts like NIT; substitution and income effects reduce work; participation effects can be clearer than hours.

Figure 3.7: Work Incentives of a Refundable Tax Credit

(a) CWB type Program



(b) CWB compared to welfare



Unemployment Insurance (EI): Key Features

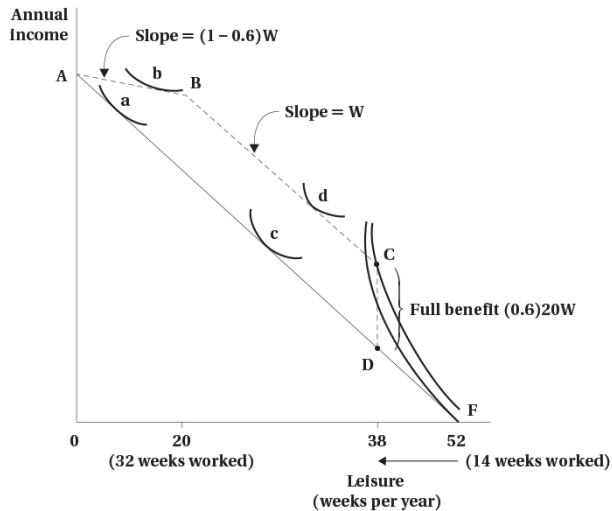
- Largest income security program for non-elderly in Canada.
- Replacement rate $\approx 55\%$ of lost earnings up to a maximum.
- Benefit duration: roughly 14–45 weeks, varying by local unemployment.
- Eligibility: roughly 420–700 hours worked (about 12–20 weeks at 35 hrs/week), depending on region.

Worked Example: Unemployment Insurance and Work Incentives

Suppose recipients receive 60% of weekly pay if unemployed; must have worked at least 14 weeks to qualify; maximum benefit duration is 20 weeks.

- **Income–leisure lens:** UI provides income when hours fall to zero, shifting choices along a non-linear budget across employment and non-employment states.
- **Predictions:** Longer/greater benefits $\Rightarrow$ longer search duration; but also better job matching and consumption smoothing.

Income Constraint for a Simplified Unemployment Insurance Scheme



- Alters budget (medical costs, fewer feasible hours) and/or preferences (higher disutility of work).
- Consider hours able to work, medical expenses, reduced wage-earning capacity, and relative disutility of market work.

Figure 3.8: Effect of Disability

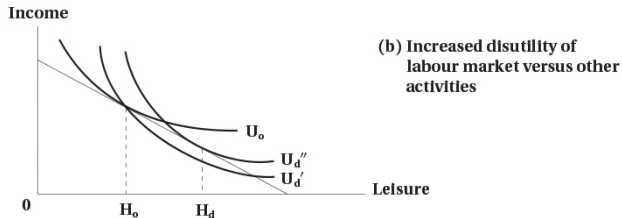
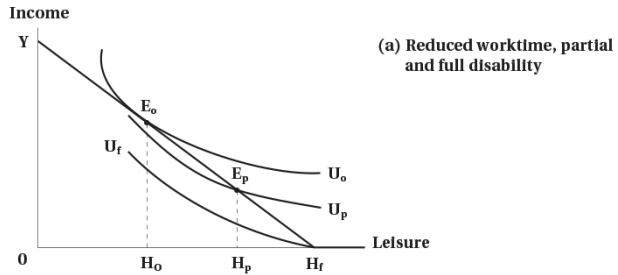
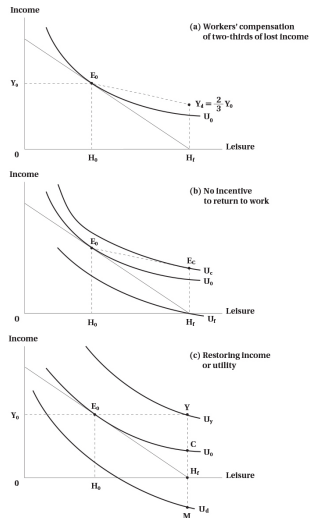
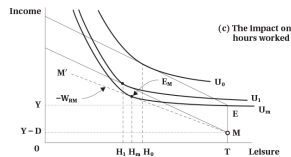
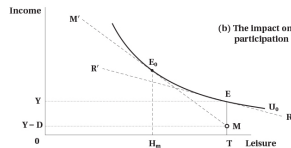
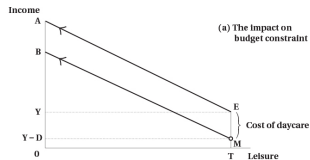


Figure 3.9: Effect of Compensation



- Daycare can be modeled as a **fixed cost of working**: reduces intercept or creates an entry “gap.”
- **Implications:** Raises reservation wage; can reduce hours for existing workers; can create non-convexities.
- Subsidies lower fixed cost, encouraging participation and part-time work; may reduce hours for some already working (income effect).

Figure 3.10: The Effect of Child Care Costs on Labour Supply



Summary (1/2)

- Income maintenance schemes top up low incomes from varied causes; some are better for transitory shocks, others for persistent low wages.
- The labour supply framework compares work incentives across designs via their budget constraints.
- **Demogrants:** Pure income effect; reduce work incentives at the margin.
- **Welfare:** Adverse incentives magnified by high clawbacks; policy levers include earnings disregards and lower implicit tax rates.

Summary (2/2)

- **NIT:** Smoother guarantee with partial clawback; still reduces work relative to no transfer.
- **Wage Subsidies/Refundable Credits:** Encourage work on phase-in but can reduce hours on flat/phase-out; participation effects often clearer.
- **UI/EI:** Insures transitory hours shocks; trade-off between search/matching and potential work disincentives.
- **Disability/Workers' Comp:** Shift constraints and preferences; evaluate well-being across program designs.
- **Childcare:** Fixed costs affect participation margins; subsidies can raise LFPR, especially for parents.

Thank You!